

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
In [2]: ipl=pd.read_csv(r"C:\Users\Tanmay\OneDrive\Desktop\ipl_2026_matches-selected-column
ipl
```

Out[2]:

	match_no	date	team1	team2	venue	city	winner	winn
0	1	2026-03-28	Royal Challengers Bengaluru	Sunrisers Hyderabad	M. Chinnaswamy Stadium	Bengaluru	Royal Challengers Bengaluru	wick
1	2	2026-03-29	Mumbai Indians	Kolkata Knight Riders	Wankhede Stadium	Mumbai	Kolkata Knight Riders	wick
2	3	2026-03-30	Chennai Super Kings	Rajasthan Royals	M. A. Chidambaram Stadium	Chennai	Rajasthan Royals	wick
3	4	2026-03-31	Gujarat Titans	Punjab Kings	Narendra Modi Stadium	Ahmedabad	Punjab Kings	wick
4	5	2026-04-01	Delhi Capitals	Lucknow Super Giants	Arun Jaitley Stadium	Delhi	Delhi Capitals	wick
...	...	...	...	...	...	...	...	...
69	70	2026-06-04	Rajasthan Royals	Sunrisers Hyderabad	Sawai Mansingh Stadium	Jaipur	Sunrisers Hyderabad	wick
70	71	2026-05-26	Royal Challengers Bengaluru	Gujarat Titans	HPCA Cricket Stadium	Dharamshala	Royal Challengers Bengaluru	wick
71	72	2026-05-27	Sunrisers Hyderabad	Rajasthan Royals	Maharaja Yadavindra Singh Stadium	Mullanpur	Rajasthan Royals	wick
72	73	2026-05-29	Rajasthan Royals	Gujarat Titans	Maharaja Yadavindra Singh Stadium	Mullanpur	Gujarat Titans	wick
73	74	2026-05-31	Gujarat Titans	Royal Challengers Bengaluru	Narendra Modi Stadium	Ahmedabad	Royal Challengers Bengaluru	wick

74 rows × 10 columns



```
In [49]: ipl.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 74 entries, 0 to 73
Data columns (total 10 columns):
#   Column          Non-Null Count  Dtype
---  -
0   match_no        74 non-null    int64
1   date            74 non-null    datetime64[us]
2   team1           74 non-null    str
3   team2           74 non-null    str
4   venue           74 non-null    str
5   city            74 non-null    str
6   winner          74 non-null    str
7   win_by          73 non-null    str
8   win_margin      74 non-null    int64
9   toss_winner     73 non-null    str
dtypes: datetime64[us](1), int64(2), str(7)
memory usage: 5.9 KB
```

Data Cleaning

```
In [46]: #Check for missing values in the dataset.
ipl.isnull().sum()
```

```
Out[46]: match_no        0
         date           0
         team1          0
         team2          0
         venue          0
         city           0
         winner         0
         win_by         1
         win_margin     0
         toss_winner    1
         dtype: int64
```

```
In [47]: #Check for duplicate records and remove them if found.
ipl.duplicated().sum()
```

```
Out[47]: np.int64(0)
```

```
In [48]: #Convert the date column into datetime format.
ipl["date"] = pd.to_datetime(ipl["date"])
```

Exploratory Data Analysis

```
In [51]: #How many matches were played in the tournament?
print(ipl["match_no"].count())
```

```
In [62]: #Which teams participated in the tournament?
teams = pd.concat([ipl["team1"],ipl["team2"]]).unique()
print(teams)

<StringArray>
['Royal Challengers Bengaluru',      'Mumbai Indians',
      'Chennai Super Kings',        'Gujarat Titans',
      'Delhi Capitals',              'Sunrisers Hyderabad',
      'Kolkata Knight Riders',      'Lucknow Super Giants',
      'Rajasthan Royals',           'Punjab Kings']
Length: 10, dtype: str
```

```
In [81]: #Which cities hosted the most matches?
ipl["city"].value_counts().head(1)
```

```
Out[81]: city
Ahmedabad      10
Name: count, dtype: int64
```

```
In [82]: #Which venues hosted the most matches?
ipl["venue"].value_counts().head(1)
```

```
Out[82]: venue
Narendra Modi Stadium      10
Name: count, dtype: int64
```

Team Performance Analysis

```
In [87]: #Which team won the most matches?
ipl["winner"].value_counts().head(1)
```

```
Out[87]: winner
Royal Challengers Bengaluru      12
Name: count, dtype: int64
```

```
In [79]: #Which team had the highest win percentage?
matches_played = pd.concat([ipl["team1"], ipl["team2"]]).value_counts()
matches_won = ipl["winner"].value_counts()
win_percentage = ((matches_won / matches_played) * 100).round(2)
print(win_percentage.sort_values(ascending=False))
```

```
winner
Royal Challengers Bengaluru      70.59
Punjab Kings                     69.23
Sunrisers Hyderabad              61.54
Delhi Capitals                   61.54
Gujarat Titans                   57.89
Rajasthan Royals                 55.56
Kolkata Knight Riders            33.33
Mumbai Indians                   30.77
Chennai Super Kings              25.00
Lucknow Super Giants             20.00
Match abandoned                  NaN
Name: count, dtype: float64
```

```
In [12]: #How many matches did each team play?
matches_played = pd.concat([ipl["team1"], ipl["team2"]]).value_counts()
print(matches_played)
```

```
Gujarat Titans          19
Rajasthan Royals        18
Royal Challengers Bengaluru  17
Kolkata Knight Riders    15
Lucknow Super Giants     15
Mumbai Indians          13
Delhi Capitals           13
Sunrisers Hyderabad     13
Punjab Kings            13
Chennai Super Kings      12
Name: count, dtype: int64
```

```
In [13]: #Which team won the most matches by runs?
ipl[ipl["win_by"] == "runs"]["winner"].value_counts().head(1)
```

```
Out[13]: winner
Sunrisers Hyderabad    6
Name: count, dtype: int64
```

```
In [14]: #Which team won the most matches by wickets?
ipl[ipl["win_by"] == "wickets"]["winner"].value_counts().head(1)
```

```
Out[14]: winner
Royal Challengers Bengaluru    9
Name: count, dtype: int64
```

Toss Analysis

```
In [15]: #Which team won the most matches ?
ipl["toss_winner"].value_counts().head(1)
```

```
Out[15]: toss_winner
Gujarat Titans    12
Name: count, dtype: int64
```

```
In [16]: #How often did the toss winner also win the match?
print((ipl["toss_winner"] == ipl["winner"]).sum())
```

54

```
In [80]: #Does winning the toss appear to provide an advantage?
advantage = ((ipl["toss_winner"] == ipl["winner"]).mean() * 100).round(2)
print("Toss winner won", advantage, "% of matches.")
```

Toss winner won 72.97 % of matches.

Match Margin Analysis

```
In [18]: #What was the biggest victory of the season ?
ipl.loc[ipl["win_margin"].idxmax()]
```

```
Out[18]: match_no          19
         date            2026-04-14 00:00:00
         team1           Chennai Super Kings
         team2           Mumbai Indians
         venue           M. A. Chidambaram Stadium
         city            Chennai
         winner           Chennai Super Kings
         win_by           runs
         win_margin       103
         toss_winner      Chennai Super Kings
         Name: 18, dtype: object
```

```
In [19]: #What is the average winning margin ?
         print(ipl["win_margin"].mean().round(2))
```

17.99

```
In [89]: #Which team recorded the largest winning margin ?
         ipl.loc[ipl["win_margin"].idxmax(), "winner"]
```

```
Out[89]: 'Chennai Super Kings'
```

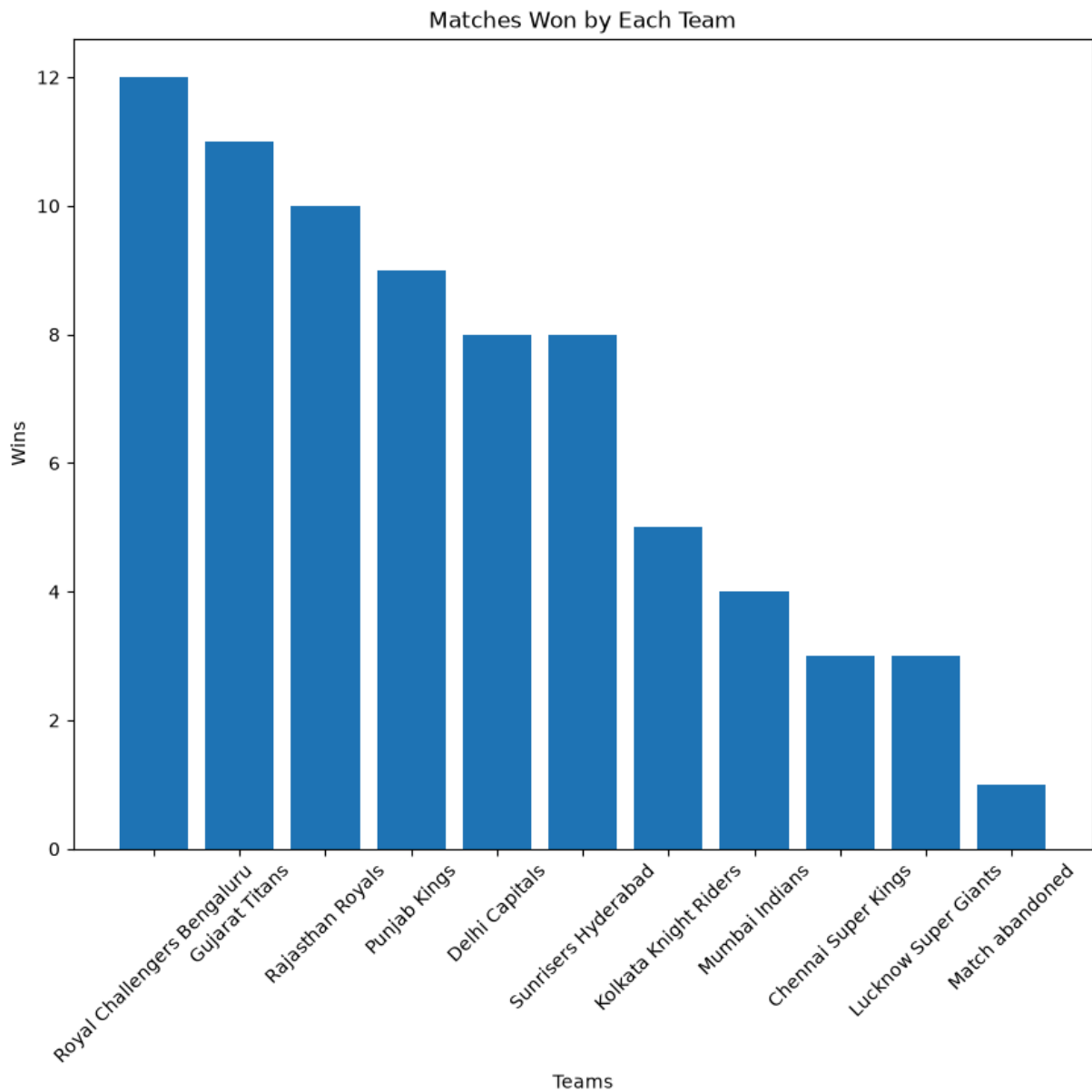
Venue & City Insights

```
In [22]: #Which venue produced the most competitive matches (smallest average winning margin)
         venue_margin = ipl.groupby("venue")["win_margin"].mean()
         print(venue_margin.sort_values().head(1))
```

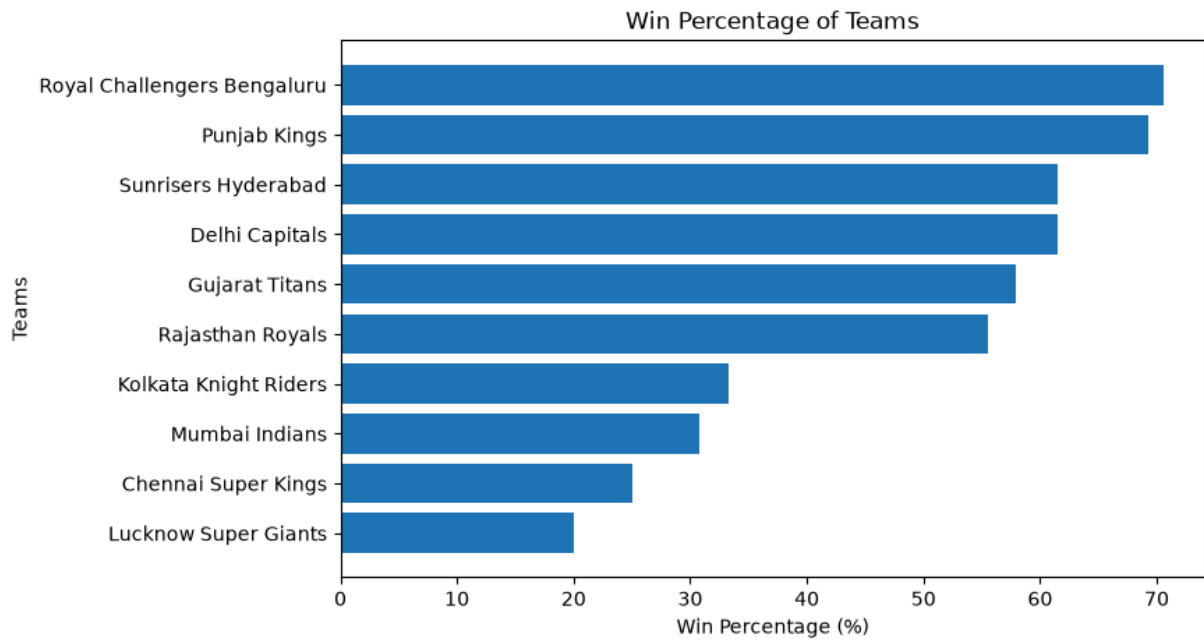
```
venue
Maharaja Yadavindra Singh Stadium    3.5
Name: win_margin, dtype: float64
```

Matplotlib Visualization Tasks

```
In [37]: #Create a bar chart showing the number of matches won by each team.
         wins = ipl["winner"].value_counts()
         plt.figure(figsize=(10,8))
         plt.bar(wins.index, wins.values)
         plt.title("Matches Won by Each Team")
         plt.xlabel("Teams")
         plt.ylabel("Wins")
         plt.xticks(rotation=45)
         plt.show()
```

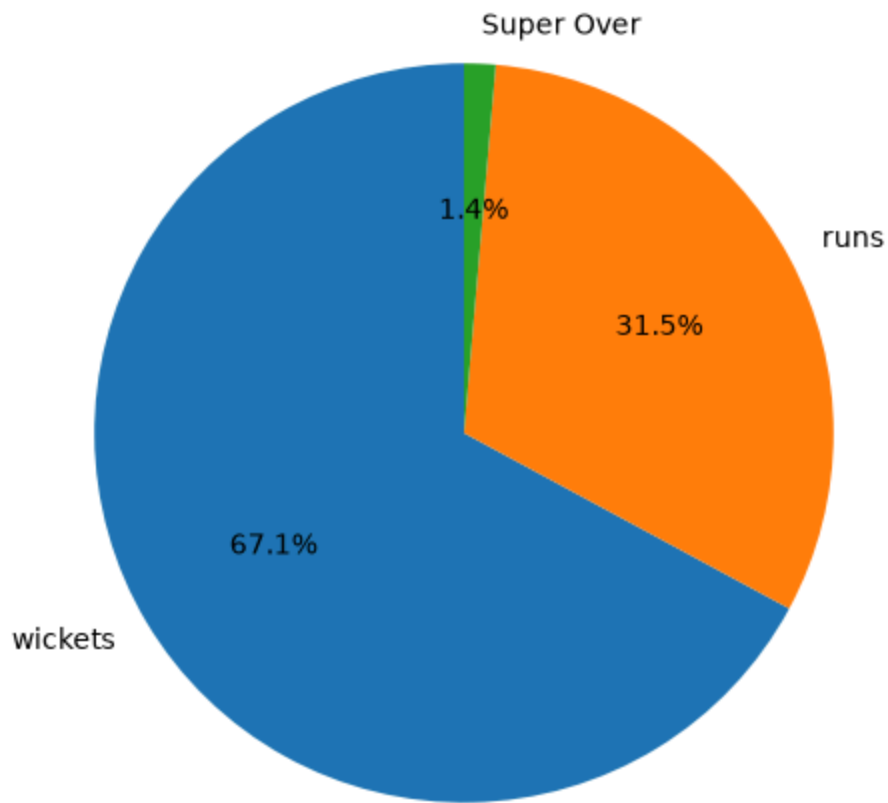


```
In [26]: #Create a horizontal bar chart showing the win percentage of every team (highest to
matches_played = pd.concat([ipl["team1"], ipl["team2"]]).value_counts()
matches_won = ipl["winner"].value_counts()
win_percentage = ((matches_won / matches_played) * 100).round(2)
win_percentage = win_percentage.sort_values()
plt.figure(figsize=(8,5))
plt.barh(win_percentage.index, win_percentage.values)
plt.title("Win Percentage of Teams")
plt.xlabel("Win Percentage (%)")
plt.ylabel("Teams")
plt.show()
```

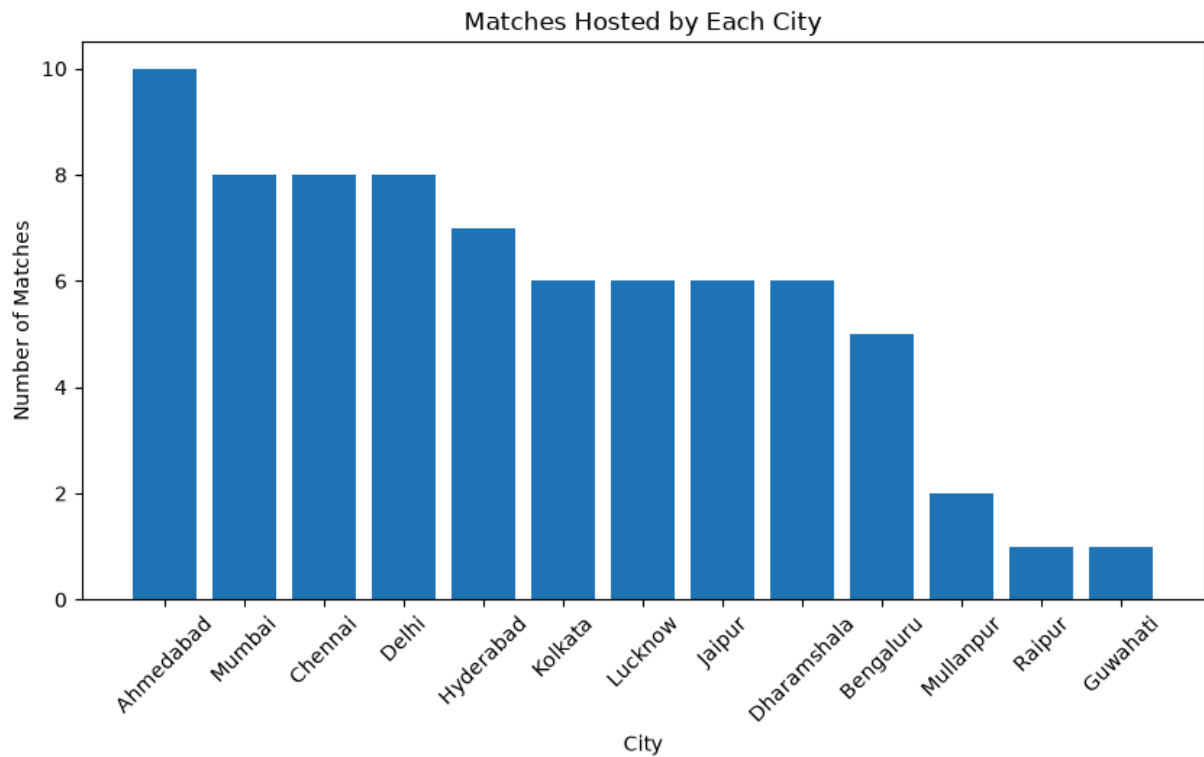


```
In [27]: #Create a pie chart showing the distribution of matches won by runs versus wickets.
win_type = ipl["win_by"].value_counts()
plt.figure(figsize=(6,6))
plt.pie(
    win_type.values,
    labels=win_type.index,
    autopct="%1.1f%%",
    startangle=90
)
plt.title("Wins by Runs vs Wickets")
plt.show()
```

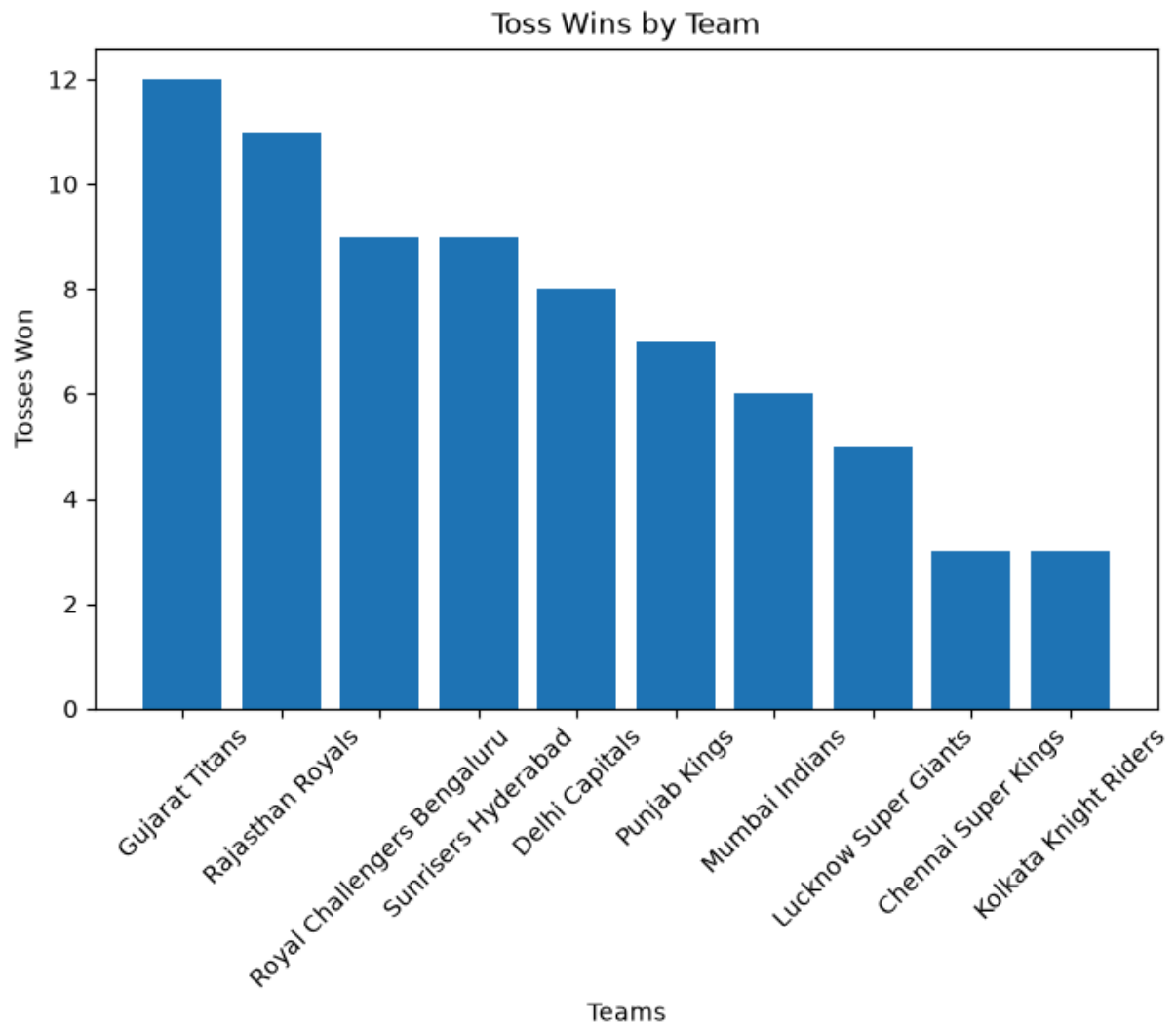
Wins by Runs vs Wickets



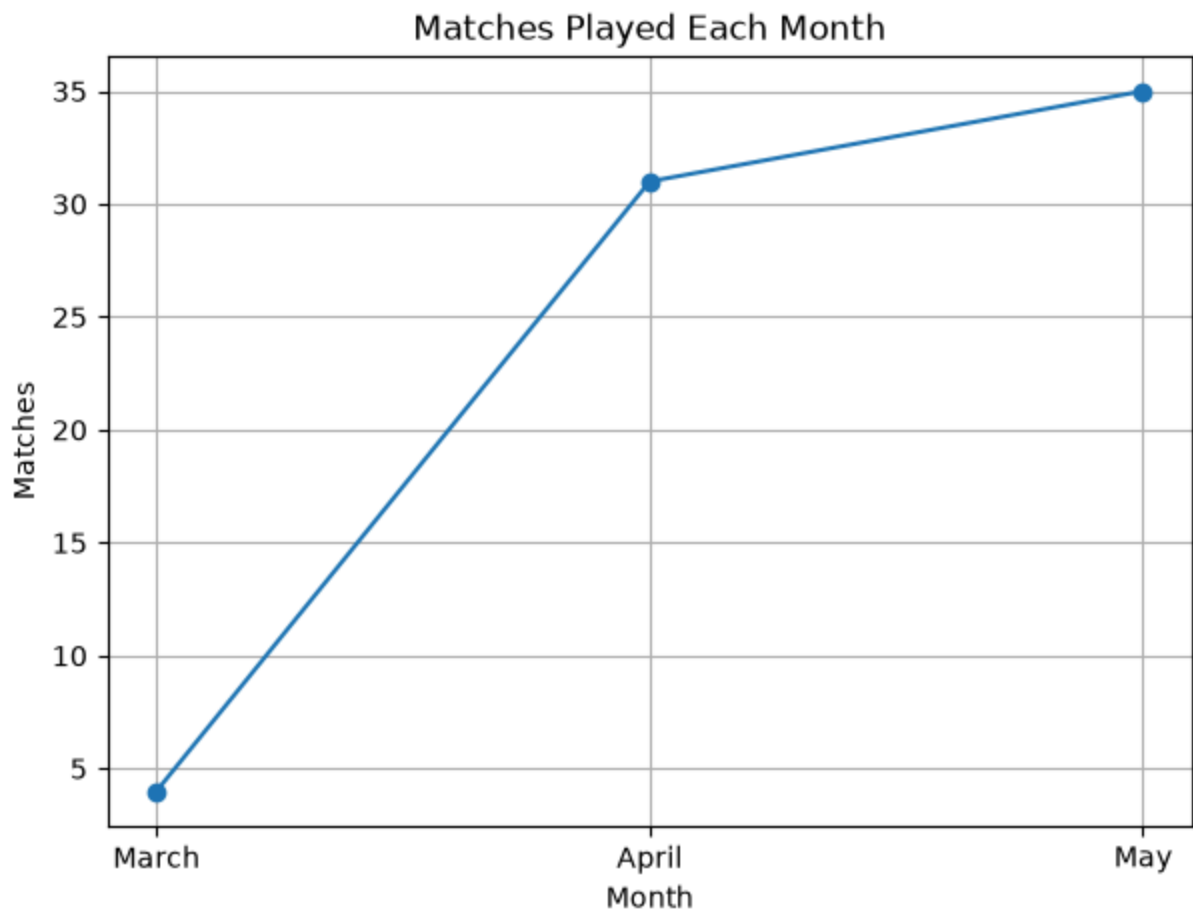
```
In [28]: #Create a bar chart showing the number of matches hosted in each city.
city_matches = ipl["city"].value_counts()
plt.figure(figsize=(10,5))
plt.bar(city_matches.index, city_matches.values)
plt.title("Matches Hosted by Each City")
plt.xlabel("City")
plt.ylabel("Number of Matches")
plt.xticks(rotation=45)
plt.show()
```

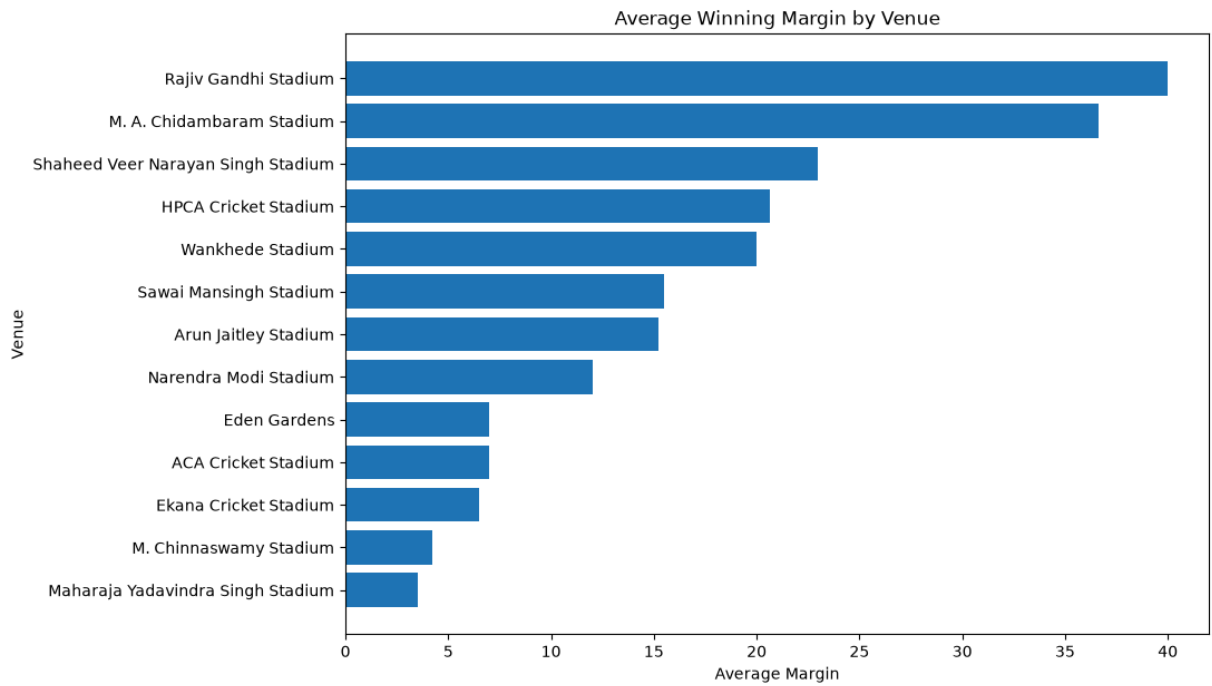
```
In [41]: #Create a bar chart showing the number of tosses won by each team.
toss = ipl["toss_winner"].value_counts()
plt.figure(figsize=(8,5))
plt.bar(toss.index, toss.values)
plt.title("Toss Wins by Team")
plt.xlabel("Teams")
plt.ylabel("Tosses Won")
plt.xticks(rotation=45)
plt.show()
```



```
In [30]: #Create a line graph showing how many matches were played each month (extract the m
ipl["date"] = pd.to_datetime(ipl["date"])
ipl["Month"] = ipl["date"].dt.month_name()
month_order = ["March", "April", "May"]
monthly_matches = (
    ipl.groupby("Month")
    .size()
    .reindex(month_order)
)
plt.figure(figsize=(7,5))
plt.plot(
    monthly_matches.index,
    monthly_matches.values,
    marker="o"
)
plt.title("Matches Played Each Month")
plt.xlabel("Month")
plt.ylabel("Matches")
plt.grid()
plt.show()
```



```
In [31]: #Create a horizontal bar chart showing the average winning margin at each venue.
venue_margin = (
    ipl.groupby("venue")["win_margin"]
    .mean()
    .sort_values()
)
plt.figure(figsize=(10,7))
plt.barh(
    venue_margin.index,
    venue_margin.values
)
plt.title("Average Winning Margin by Venue")
plt.xlabel("Average Margin")
plt.ylabel("Venue")
plt.show()
```



```
In [32]: #Create a scatter plot to examine the relationship between match number (match_no)
plt.figure(figsize=(8,5))
plt.scatter(
    ipl["match_no"],
    ipl["win_margin"]
)
plt.title("Match Number vs Winning Margin")
plt.xlabel("Match Number")
plt.ylabel("Winning Margin")
plt.grid()
plt.show()
```

